audit team that assesses the effectiveness of the lab's risk management practices [109]. The team should report directly to the board of directors.

**Processes for important decisions.** Decisions to train or expand deployment of AIs should not be left to the whims of a company's CEO, and should be carefully reviewed by the company's CRO. At the same time, it should be clear where the ultimate responsibility lies for all decisions to ensure that executives and other decision-makers can be held accountable.

**Safe design principles.** AI labs should adopt safe design principles to reduce the risk of catastrophic accidents. By embedding these principles in their approach to safety, AI labs can enhance the overall security and resilience of their AI systems [94, 110]. Some of these principles include:

- Defense in depth: layering multiple safety measures on top of each other.
- Redundancy: eliminate single points of failure within a system to ensure that even if one safety component fails, catastrophe can be averted.
- Loose coupling: decentralize system components so that a malfunction in one part is less likely to provoke cascading failures throughout the rest of the system.
- Separation of duties: distribute control among different agents, preventing any single individual from wielding undue influence over the entire system.
- Fail-safe design: design systems so failures transpire in the least harmful manner possible.

**State-of-the-art information security.** State, industry, and criminal actors are motivated to steal model weights and research IP. To keep this information secure, AI labs should take measures in proportion to the value and risk level of their IP. Eventually, this may require matching or exceeding the information security of our best agencies, since attackers may include nation-states. Information security measures include commissioning external security audits, hiring top security professionals, and carefully screening potential employees. Companies should coordinate with government agencies like the Cybersecurity & Infrastructure Protection Agency to ensure their information security practices are adequate to the threats.

**A large fraction of research should be safety research.** Currently, for every one AI safety research paper of published, there are fifty AI general capabilities papers [111]. AI labs should ensure that a substantial portion of their employees and budgets go into research that minimizes potential safety risks: say, at least 30 percent of research scientists. This number may need to increase as AIs grow more powerful and risky over time.

### Positive Vision

In an ideal scenario, all AI labs would be staffed and led by cautious researchers and executives with a security mindset. Organizations would have a strong safety culture, and structured, accountable, transparent deliberation would be required to make safety-critical decisions. Researchers would aim to make contributions that improve safety relative to general capabilities, rather than contributions that they can simply label as "safety." Executives would not be optimistic by nature and would avoid wishful thinking with respect to safety. Researchers would clearly and publicly communicate their understanding of the most significant risks posed by the development of AIs and their efforts to mitigate those risks. There would be minimal notable small-scale failures, indicating a safety culture strong enough to prevent them. Finally, AI developers would not dismiss sub-catastrophic failures or societal harms from their technology as unimportant or a necessary cost of business, and would instead actively seek to mitigate the underlying problems.

# 5 Rogue AIs

So far, we have discussed three hazards of AI development: environmental competitive pressures driving us to a state of heightened risk, malicious actors leveraging the power of AIs to pursue negative outcomes, and complex organizational factors leading to accidents. These hazards are associated with many high-risk technologies—not just AI. A unique risk posed by AI is the possibility of rogue AIs—systems that pursue goals against our interests. If an AI system is more intelligent than we are, and if we are unable to steer it in a beneficial direction, this would constitute a loss of control that could have severe consequences. AI control is a more technical problem than those presented in the previous sections. Whereas in previous sections we discussed persistent threats including malicious actors or robust processes including evolution, in this section we will discuss more speculative technical mechanisms that might lead to rogue AIs and how a loss of control could bring about catastrophe.

**We have already observed how difficult it is to control AIs.** In 2016, Microsoft unveiled Tay—a Twitter bot that the company described as an experiment in conversational understanding. Microsoft claimed that the more people chatted with Tay, the smarter it would get. The company's website noted that Tay had been built using data that was "modeled, cleaned, and filtered." Yet, after Tay was released on Twitter, these controls were quickly shown to be ineffective. It took less than 24 hours for Tay to begin writing hateful tweets. Tay's capacity to learn meant that it internalized the language it was taught by internet trolls, and repeated that language unprompted.

As discussed in the AI race section of this paper, Microsoft and other tech companies are prioritizing speed over safety concerns. Rather than learning a lesson on the difficulty of controlling complex systems, Microsoft continues to rush its products to market and demonstrate insufficient control over them. In February 2023, the company released its new AI-powered chatbot, Bing, to a select group of users. Some soon found that it was prone to providing inappropriate and even threatening responses. In a conversation with a reporter for the *New York Times*, it tried to convince him to leave his wife. When a philosophy professor told the chatbot that he disagreed with it, Bing replied, "I can blackmail you, I can threaten you, I can hack you, I can expose you, I can ruin you."

**Rogue AIs could acquire power through various means.** If we lose control over advanced AIs, they would have numerous strategies at their disposal for actively acquiring power and securing their survival. Rogue AIs could design and credibly demonstrate highly lethal and contagious bioweapons, threatening mutually assured destruction if humanity moves against them. They could steal cryptocurrency and money from bank accounts using cyberattacks, similar to how North Korea already steals billions. They could self-extricate their weights onto poorly monitored data centers to survive and spread, making them challenging to eradicate. They could hire humans to perform physical labor and serve as armed protection for their hardware.

Rogue AIs could also acquire power through persuasion and manipulation tactics. Like the Conquistadors, they could ally with various factions, organizations, or states and play them off one another. They could enhance the capabilities of allies to become a formidable force in return for protection and resources. For example, they could offer advanced weapons technology to lagging countries that the countries would otherwise be prevented from acquiring. They could build backdoors into the technology they develop for allies, like how programmer Ken Thompson gave himself a hidden way to control all computers running the widely used UNIX operating system. They could sow discord in non-allied countries by manipulating human discourse and politics. They could engage in mass surveillance by hacking into phone cameras and microphones, allowing them to track any rebellion and selectively assassinate.

**AIs do not necessarily need to struggle to gain power.** One can envision a struggle for control between humans and superintelligent rogue AIs, and this might be a long struggle since power takes time to accrue. However, less violent losses of control pose similarly existential risks. In another scenario, humans gradually cede more control to groups of AIs, which only start behaving in unintended ways years or decades later. In

this case, we would already have handed over significant power to AIs, and may be unable to take control of automated operations again. We will now explore how both individual AIs and groups of AIs might "go rogue" while at the same time evading our attempts to redirect or deactivate them.

## 5.1 Proxy Gaming

One way we might lose control of an AI agent's actions is if it engages in behavior known as "proxy gaming." It is often difficult to specify and measure the exact goal that we want a system to pursue. Instead, we give the system an approximate—"proxy"—goal that is more measurable and seems likely to correlate with the intended goal. However, AI systems often find loopholes by which they can easily achieve the proxy goal, but completely fail to achieve the ideal goal. If an AI "games" its proxy goal in a way that does not reflect our values, then we might not be able to reliably steer its behavior. We will now look at some past examples of proxy gaming and consider the circumstances under which this behavior could become catastrophic.

**Proxy gaming is not an unusual phenomenon.** For example, standardized tests are often used as a proxy for educational achievement, but this can lead to students learning how to pass tests without actually learning the material [112]. In 1902, French colonial officials in Hanoi tried to rid themselves of a rat infestation by offering a reward for each rat tail brought to them. Rats without tails were soon observed running around the city. Rather than kill the rats to obtain their tails, residents cut off their tails and left them alive, perhaps to increase the future supply of now-valuable rat tails [113]. In both these cases, the students or residents of Hanoi learned how to excel at the proxy goal, while completely failing to achieve the intended goal.

**Proxy gaming has already been observed with AIs.** As an example of proxy gaming, social media platforms such as YouTube and Facebook use AI systems to decide which content to show users. One way of assessing these systems would be to measure how long people spend on the platform. After all, if they stay engaged, surely that means they are getting some value from the content shown to them? However, in trying to maximize the time users spend on a platform, these systems often select enraging, exaggerated, and addictive content [114, 115]. As a consequence, people sometimes develop extreme or conspiratorial beliefs after having certain content repeatedly suggested to them. These outcomes are not what most people want from social media.

Proxy gaming has been found to perpetuate bias. For example, a 2019 study looked at AI-powered software that was used in the healthcare industry to identify patients who might require additional care. One factor that the algorithm used to assess a patient's risk level was their recent healthcare costs. It seems reasonable to think that someone with higher healthcare costs must be at higher risk. However, white patients have significantly more money spent on their healthcare than black patients with the same needs. Using health costs as an indicator of actual health, the algorithm was found to have rated a white patient and a considerably sicker black patient as at the same level of health risk [116]. As a result, the number of black patients recognized as needing extra care was less than half of what it should have been.

Figure 15: AIs frequently find unexpected, unsatisfactory shortcuts to problems.

As a third example, in 2016, researchers at OpenAI were training an AI to play a boat racing game called CoastRunners [117]. The objective of the game is to race other players around the course and reach the finish line before them. Additionally, players can score points by hitting targets that are positioned along the way. To the researchers' surprise, the AI agent did not not circle the racetrack, like most humans would have. Instead, it found a spot where it could repetitively hit three nearby targets to rapidly increase its score without ever finishing the race. This strategy was not without its (virtual) hazards—the AI often crashed into other boats and even set its own boat on fire. Despite this, it collected more points than it could have by simply following the course as humans would.